SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – INDIVIDUAL TALK

Course Code:	ITA03	**Course Title:**	Mobile computing
CO Assessed:	CO1	**Assessment:**	Assessment Tool 3 – INDIVIDUAL TALK
Weightage:	30 % of CIA	**Total Marks:**	20 marks
CO1 Statement:	*Analyse the mobile network components, mobile base station, mobile infrastructures and protocols in cellular systems.*		
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Section / Batch:	2023	**Date:**	23/10/2026

Code of Conduct

I __Narne Tejaswi (192372120)_____(Name / Reg. No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Instructions

- Topic given before the session and students speak on stage.
- Duration: 10 minutes (In class)

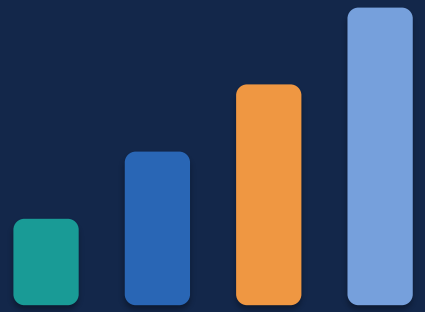
Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

1. Performance Metrics of Cellular Networks

PERFORMANCE METRICS OF CELLULAR NETWORKS



Understanding how we measure mobile network quality



Narne Tejaswi

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Network Performance Metrics

Simple measures that tell us how well a mobile network is working.

Cellular networks such as 4G and 5G connect our phones to the internet and to each other. Performance metrics help us check whether the connection is fast, stable, and reliable.

Speed

How quickly data is sent or received. Example: downloading a video.

Delay

How long data takes to travel across the network. Important for video calls and gaming.

Reliability

How consistently the connection works without drops or interruptions.

Why it matters: better measurements help operators improve coverage, capacity, and user experience.

Key Metrics Used to Measure Performance

Each metric explains a different part of the mobile experience.

Throughput

Actual data speed received by the user, usually measured in Mbps.

Latency

Time taken for data to travel; lower latency feels more responsive.

Jitter

Variation in delay; low jitter helps calls and live streaming sound smooth.

Packet Loss

Data packets that fail to reach their destination; lower is better.

Signal Strength

How strong the radio signal is between the phone and cell tower.

Handover Success

Whether a call or data session continues while moving between cells.

Good performance usually means high throughput and signal quality, with low latency, jitter, and packet loss.

How These Metrics Affect Daily Use

The same network can feel different depending on what you are doing.

Video Streaming

Needs enough throughput to play clearly without buffering.

Video Calls

Need low latency, low jitter, and very little packet loss.

Online Gaming

Benefits from low delay and a stable connection.

Travelling

Good handovers help calls and internet sessions continue between cell towers.

Example: a phone may show strong signal but still feel slow if the cell tower is crowded.

How Can We Measure and Improve It?

A simple process used by network engineers and service providers.

1. **Collect** — Run speed tests and record signal, delay, and dropped connections.
2. **Compare** — Check results across locations, times, and 4G/5G networks.
3. **Find the cause** — Look for weak coverage, tower congestion, interference, or device issues.
4. **Improve** — Adjust tower settings, add capacity, improve coverage, or optimize handovers.

Useful tools include mobile speed-test apps, network monitoring systems, and drive-test tools.

Conclusion

Performance metrics help us understand whether a cellular network is fast, responsive, stable, and reliable.

Measure

Speed, delay, signal, and packet loss.

Analyze

Compare results across places and times.

Improve

Use findings to make mobile service better.

Thank You!


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